Wind Turbine Project: Part II Modelling of Waves at Bretagne Sud 1 Site

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I. Introduction

A. Context and Motivation

OFFSHORE floating wind projects depend on an accurate description of the local wave environment because waves control platform motions, mooring loads, access windows, and ultimate survival. At the Bretagne Sud 1 implantation zone, the wave climate is shaped both by the large-scale North Atlantic exposure and by local bathymetric features such as the continental shelf break and the surrounding islands. Design decisions that rely only on generic deep-water conditions risk underestimating or misplacing the loads that arise after waves have refracted, shoaled, and diffracted on their way to the turbines.

Part I of this project characterised the mean and extreme wave climate at an offshore ResourceCode hindcast node representative of Bretagne Sud 1, using multi-decadal statistics, comparison with a nearby buoy, and univariate and multivariate extreme value analyses [1]. That study established that the site experiences predominantly deep-water waves with a westerly incidence, quantified 1- and 50-year design heights, and constructed an environmental contour in significant wave height and peak period. These results provide robust offshore design conditions but do not yet describe how those sea states are transformed as they propagate across the shelf or how they translate into forces on individual support structures.

The present report addresses this gap by introducing explicit spatial and structural modelling between the offshore climate and local wave loads. The focus is on how long-period storms propagating from the west–south–west interact with the complex bathymetry around southern Brittany and how that interaction modifies wave heights and directions at the turbine site. In parallel, laboratory measurements of wave-induced forces on a vertical cylinder are combined with linear diffraction theory to infer prototype monopile loads. This modelling chain, from hindcast extremes through wave transformation to structural response, is central for assessing the viability and safety of a floating wind farm at Bretagne Sud 1.

B. Statement of Purpose

The purpose of this report is to use numerical and experimental modelling of waves to bridge the gap between offshore hindcast statistics and local wave-induced loads on an idealised support structure. Building on the offshore climate and extreme value analysis from Part I, satellite observations, geometric-optics ray tracing, and laboratory wave-tank data are combined to obtain a site-specific description of design sea states and the corresponding monopile-scale forces.

The analysis addresses three connected questions. First, how do satellite images of surface wave patterns near Bretagne Sud 1 constrain the dominant wavelength, propagation direction, and evidence of refraction and diffraction, and are these observations consistent with the hindcast-based peak periods and directions? Second, given representative 1- and 50-year storm conditions from the environmental contour, how do refraction and shoaling across the local bathymetry redistribute wave energy and modify significant wave height at the project site when modelled with a ray-tracing scheme? Third, for a regular wave representative of the transformed sea state, what horizontal forces act on a monopile-like vertical cylinder, as inferred by scaling laboratory force measurements and by applying MacCamy–Fuchs long-wave diffraction theory [2]? Answering these questions provides a physically transparent path from offshore climate statistics to local design loads.

C. Report Outline

The report is structured to move from wave climate description to local transformation and structural loading. Section II presents the satellite imagery analysis, Section III describes and applies the wave ray-tracing simulations on the regional bathymetry, and Section IV uses laboratory and theoretical models to estimate wave-induced loads on a vertical cylinder representative of a monopile. Section V concludes the report.

II. Satellite Imagery Analysis

This section uses satellite imagery to observe real wave patterns near the Bretagne Sud 1 site and to extract quantitative wave properties that can be compared with the ResourceCode hindcast data. The focus is on a single Sentinel-2 overpass that captures long-crested waves propagating across the outer shelf, from which wavelength and propagation direction are estimated, and bathymetric refraction and diffraction are analysed qualitatively.

A. Satellite Data Selection and Study Area

The satellite-based analysis uses a Sentinel-2 Level-2A image acquired on 25 March 2020, accessed through the Copernicus Data Space Ecosystem Browser [3]. The objective in this subsection is to document the satellite scene and to place the Bretagne Sud 1 site within its regional geographic setting before any quantitative measurements of wave properties are made. The study area is centred on the Bretagne Sud 1 site at $(47.32°\mathrm{N}, 3.55°\mathrm{W})$, and further details about the site and the associated ResourceCode hindcast point are provided in the preceding report on wave dynamics [1]. The date 25 March 2020 was selected because the Sentinel-2 overpass covers the area of interest, the wave crests were relatively visible, and because the scene contains few clouds.

To visualise the large-scale context, the Sentinel-2 scene is displayed at a regional scale in figure 1. The true-colour Level-2A product for 25 March 2020 was selected in the browser, the view was centred on Bretagne Sud 1 site, and a zoom level that shows both the site and the nearby coastline was chosen. This provides information on the relative position of the site of interest, the main islands, and the embayments along the southern Brittany coast, and confirms that the satellite swath extends far enough offshore to include the project area.

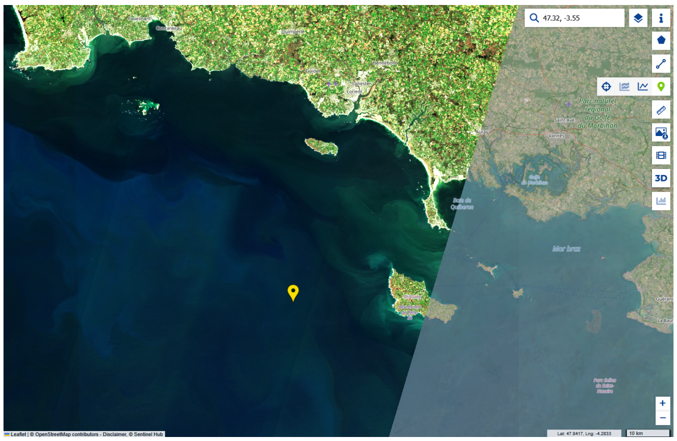

Figure 1: Sentinel-2 image of the Bretagne Sud region on 25 March 2020, viewed in the Copernicus Data Space Browser. The yellow marker indicates the approximate position of the Bretagne Sud 1 hindcast point at $(47.32°\mathrm{N}, 3.55°\mathrm{W})$. The image is shown at a regional scale to highlight the arrangement of islands, headlands, and shelf seas surrounding the site. The greyed-out area in the east was not covered by the satellite on that day.

In figure 1, the coastline, major islands, and the outer-shelf region around the yellow marker are clearly visible, whereas individual wave crests cannot be resolved at this scale. The figure shows that Bretagne Sud 1 lies well offshore, to the west of Belle-Ile-en-Mer, in a broad corridor through which Atlantic swell must pass before reaching the southern Brittany coast. This configuration implies that the wave climate at the site is influenced by both open-ocean conditions and by refraction and shadowing effects associated with the surrounding islands, which motivates the more detailed inspection of wave patterns on zoomed-in images in later subsections.

For 25 March 2020, the ResourceCode hindcast provides 24 hourly sea states at the grid node associated with Bretagne Sud 1. From these records, the daily mean and standard deviation are computed for the significant wave height H_s, the peak period T_p, the coming-from mean wave direction θ_RC (expressed in degrees clockwise from north), and the local water depth h. The resulting values are $H_\mathrm{s} = 1.95 \pm 0.15\,\mathrm{m}$, $T_\mathrm{p} = 13.39 \pm 0.71\,\mathrm{s}$, $\theta_\mathrm{RC} = 286.7 \pm 2.2°$, and $h = 93.31 \pm 1.29\,\mathrm{m}$.

From this, a reference deep-water wavelength L_0 corresponding to a wave period T is obtained from the linear deep-water dispersion relation

$$L_0(T) = \frac{gT^2}{2\pi},$$

where $g = 9.81\,\mathrm{m\,s^{-2}}$ is gravitational acceleration [4]. The wave period is taken as T_p, since it characterises the dominant swell component observed in the satellite image and so serves as a better comparison. Treating T_p as a random variable over the 24 hours gives a mean deep-water wavelength $\overline{L_0} \approx 281\,\mathrm{m}$ with standard deviation $\sigma_{L_0} \approx 30\,\mathrm{m}$. Solving the full dispersion relation at the mean depth h for $T_\mathrm{p} = 13.39\,\mathrm{s}$ yields an intermediate-water theoretical wavelength $L_\mathrm{theor} \approx 272\,\mathrm{m}$, corresponding to $h/L_\mathrm{theor} \approx 0.34$.

B. *Qualitative Description of Wave Patterns*

The qualitative description of the wave field seeks to identify how incoming swell interacts with islands and headlands in the vicinity of Bretagne Sud 1, before any numerical estimates of wavelength or direction are made. The main question is whether the regional bathymetry produces observable refraction and diffraction patterns in the same sector as the project site, which would support the later use of a ray-tracing model to describe wave propagation across the shelf.

To address this question, a more highly zoomed Sentinel-2 subscene is extracted around the northern coast of Belle-Ile-en-Mer, located east of Bretagne Sud 1. A window in the Copernicus browser that includes both offshore waters and the nearshore zone north of the island was selected, and the gain and gamma values were adjusted so that quasi-parallel bright and dark bands associated with the surface wave field became visible. Figure 2 shows how the crest geometry changes between offshore and nearshore regions.

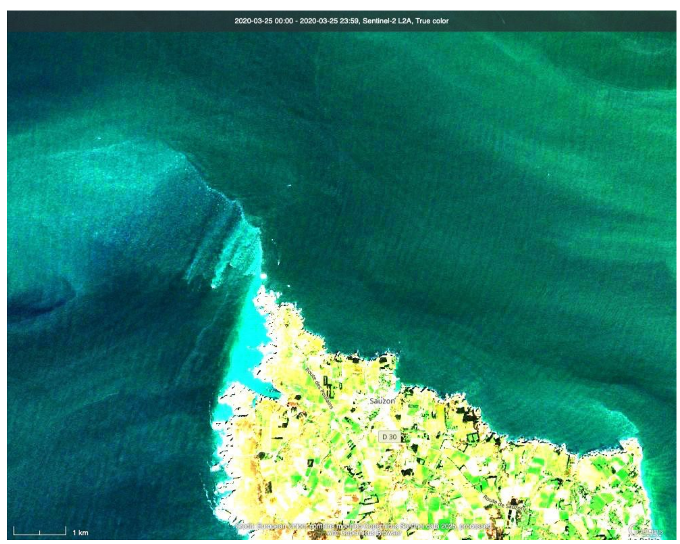

Figure 2: Sentinel-2 image on 25 March 2020 focusing on the northern coast of Belle-Ile-en-Mer, east of Bretagne Sud 1. Offshore of the island, faint, gently curved bands are visible, while nearshore the crests wrap around the headland and converge toward the coast, with a zone of reduced contrast in the lee. The scale bar at the lower left indicates 1 km.

In figure 2, the offshore part of the image shows a set of broad, nearly parallel bands extending across the shelf from west–north–west to east–south–east. Closer to the island, these bands bend around the north-western tip, become more closely spaced along the north coast, and then weaken in contrast in the shadowed area immediately leeward of the headland. These patterns are interpreted as the surface imprint of long-crested gravity waves, whose phase speed decreases as the depth shoals toward the island, causing refraction and convergence of wave rays along the north coast. Meanwhile, diffraction at the headland redistributes energy into the geometric shadow. The presence of such focusing and shadowing in the same general incident-direction sector that affects Bretagne Sud 1 indicates that bathymetric control on swell propagation is strong in this region, which justifies the more detailed, quantitative analysis of crest spacing and direction in subsequent subsections.

C. Quantitative Wave Properties from Satellite Imagery

The quantitative analysis extracts two directly measurable properties from the Sentinel-2 image in the offshore region near Bretagne Sud 1: the dominant wavelength L_{obs} and the coming-from direction θ_{obs} of the wave field. Both are estimated using the Copernicus browser tools at the native $10\,\mathrm{m}$ resolution, assuming that the visible pattern corresponds to the dominant peak of the wave spectrum and that small-scale intensity variations along a crest do not strongly bias the measurements.

1) Observed Wavelength: Wavelength is obtained by measuring crest-to-crest distances along a transect approximately aligned with the local direction of wave propagation. In the Copernicus browser, a polyline is drawn perpendicular to several successive crests, and the distances between successive bright bands are read from the *Measure distance* tool.

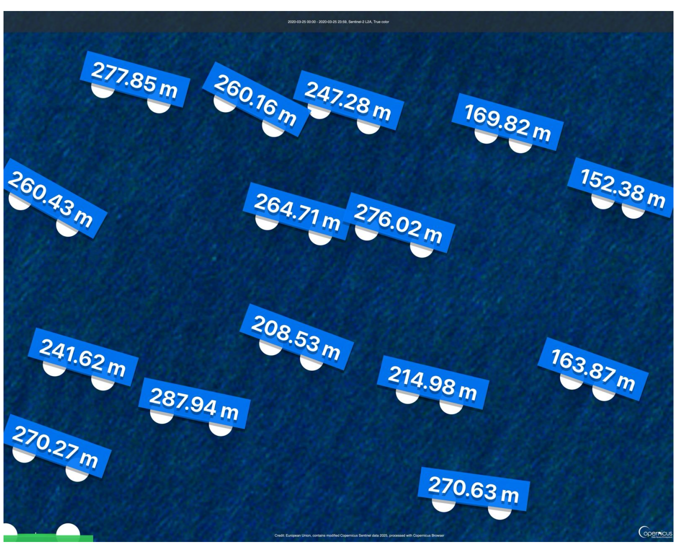

Figure 3: Detail of the Sentinel-2 image on 25 March 2020 showing a measurement transect (white line) perpendicular to the wave crests offshore of Bretagne Sud 1. The blue labels give individual crest-to-crest distances used to estimate the observed wavelength L_{obs}.

For the transect in figure 3, 15 crest-to-crest distances between approximately $150\,\mathrm{m}$ and $290\,\mathrm{m}$ were recorded. Their sample mean and standard deviation are

$$L_{\mathrm{obs}} \approx 238\,\mathrm{m}, \qquad \sigma_L \approx 45\,\mathrm{m},$$

so the dominant wavelength along this transect is about 240 with roughly $20\,\%$ variability along the sampled segment. The standard error on the mean is about $12\,\mathrm{m}$, which reflects both natural modulation of the wave field and measurement uncertainty associated with manual placement of the markers and the $10\,\mathrm{m}$ pixel size.

Comparing this with the theoretical wavelength from the hindcast, the linear dispersion relation at $h = 93.31\,\mathrm{m}$ and $T_{\mathrm{p}} = 13.3908\,\mathrm{s}$ gives $L_{\mathrm{theor}} \approx 272\,\mathrm{m}$, while the deep-water reference based on the daily T_{p} statistics is $\overline{L_0} \approx 281 \pm 30\,\mathrm{m}$. The satellite estimate is therefore shorter than L_{theor} by about $34\,\mathrm{m}$, or roughly $12\,\%$. Given the combined spread of the image-derived wavelengths ($\sigma_L \approx 45\,\mathrm{m}$) and the variability of T_{p} over the day ($\sigma_{L_0} \approx 30\,\mathrm{m}$), the intervals $L_{\mathrm{obs}} \pm \sigma_L$ and $L_{\mathrm{theor}} \pm \sigma_{L_0}$ overlap, which indicates that the observed crest spacing is broadly consistent with the hindcast peak period and linear wave theory. Residual differences are plausibly associated with local depth variations along the transect, directional and spectral spreading, and the fact that the daily-mean T_{p} is only a proxy for the exact period at the overpass time.

2) Wave Propagation Direction and Crest Orientation: Wave propagation direction is inferred from the orientation of individual crests relative to geographic north. In the Copernicus overlay, the angle α of each crest segment is measured and then converted to a coming-from direction θ_{obs} by assuming that the crests are perpendicular to the direction of travel and applying a $90°$ rotation in the clockwise-from-north convention.

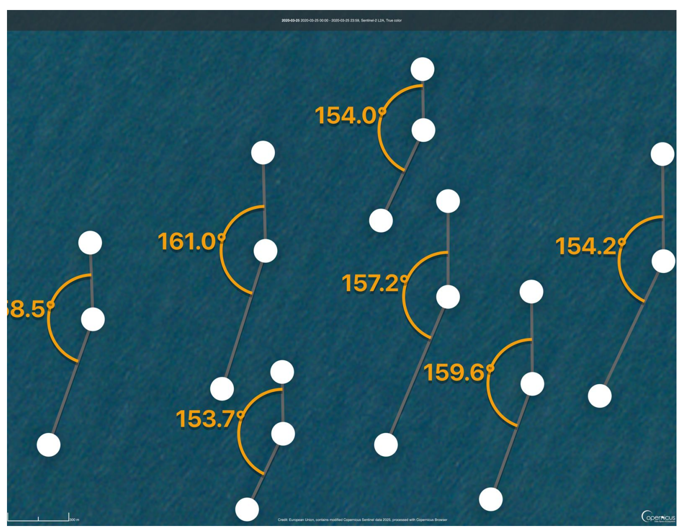

Figure 4: Close-up of wave crests offshore of Bretagne Sud 1 on 25 March 2020. The grey segments approximate local crest orientation, and the orange arcs and labels show the measured crest angles α in the Copernicus overlay. These angles are converted to coming-from directions θ_{obs} in the oceanographic convention. The cropped value on the left is $158.5°$

From figure 4, 7 crest angles were read, including the cropped value on the left, which is $158.5°$. The mean angle was calculated to be $\overline{\alpha} \approx 156.9°$ with standard deviation $\sigma_\alpha \approx 3.0°$. The crests are therefore nearly parallel, consistent with a well-organised swell field.

The coming-from direction is taken as the crest normal in a clockwise-from-north convention,

$$\theta_{\text{obs}} = (90^\circ - \alpha) \bmod 360^\circ,$$

which yields a coming-from direction with mean $\overline{\theta}_{\text{obs}} \approx 293.1^\circ$ and standard deviation $\sigma_\theta \approx 3.0^\circ$. The waves visible in the Sentinel-2 image thus arrive from the west–north-west, with only modest directional scatter along the sampled crests.

The daily mean ResourceCode hindcast direction is $\theta_{\text{RC}} = 286.73^\circ \pm 2.24^\circ$, so the satellite-based estimate is about 6.4° more northerly. This offset is comparable to the combined uncertainties in the image measurements and hindcast statistics and can be attributed to any combination of errors from manual measurement, local refraction between the hindcast node and the satellite measurement area, temporal changes between individual hours within the day, and contributions from secondary wave systems that affect the visible crests more strongly than the bulk directional statistics.

D. Linking Satellite Observations to ResourceCode Hindcast

The consistency between satellite-derived and hindcast properties is summarised in table I. The wavelength row compares the observed crest spacing with the theoretical ResourceCode value from the dispersion relation at the mean depth and peak period, while the direction row compares the observed coming-from direction with the daily mean hindcast direction. Refraction and diffraction signatures are summarised qualitatively because no systematic measurements were made in figure 2.

Table I: Comparison between satellite-derived wave properties and ResourceCode hindcast quantities for 25 March 2020 near the Bretagne Sud 1 site. Satellite and hindcast values are given with one-standard-deviation uncertainties.

Quantity	Satellite	ResourceCode
Wavelength L (m)	238 ± 45	272 ± 30
Direction θ (deg, coming-from)	293.1 ± 3.0	286.7 ± 2.2

Table I shows that both wavelength and direction from the satellite snapshot fall within the combined uncertainty envelopes of the hindcast-based theoretical values. The moderate underestimation of wavelength and the slightly more northerly direction in the image are consistent with expected spatial and temporal variability, as well as with local refraction over the shelf and around nearby islands such as Belle-Ile-en-Mer.

E. Uncertainties and Limitations

The satellite-derived wave properties are subject to several sources of uncertainty. The $10\,\text{m}$ spatial resolution of Sentinel-2 limits the precision of distance and angle measurements, especially when crests are oblique to the pixel grid. Manual placement of measurement markers introduces additional scatter, and only a finite number of crests are visible with sufficient contrast for analysis. Off-nadir viewing geometry and residual atmospheric effects may further distort apparent crest positions.

Conceptual limitations arise from approximating the visible pattern as a single, monochromatic wave system. In reality, the sea state contains a spectrum of frequencies and directions, and the hindcast represents this spectrum through bulk parameters such as H_{s}, T_{p}, and mean direction. The assumption that the sea state is quasi-stationary over the day justifies the use of daily-mean statistics for context, but any short-lived events or rapid directional changes between hourly records and the exact overpass time are not captured.

Despite these limitations, the Sentinel-2 imagery provides an independent, spatially resolved view of the wave field that complements the pointwise hindcast data. The agreement between image-based and hindcast-derived wavelengths and directions supports the use of ResourceCode extremes as input to the ray-tracing simulations and illustrates how regional bathymetry, exemplified by Belle-Ile-en-Mer, can refract and diffract incoming swell in the sector that influences the Bretagne Sud 1 site.

III. Wave Ray-Tracing Simulations

This section uses a geometric-optics ray-tracing model to propagate the extreme offshore sea states from the hindcast at Bretagne Sud 1 across the high-resolution bathymetry [5]. The objective is to quantify how refraction and shoaling modify the wave energy that reaches the project site, to obtain design wave heights that are consistent with both the large-scale extreme value analysis and the local seabed geometry used later in the monopile load calculations.

A. Input Wave Conditions

1) Methodology: The offshore wave forcing for the ray-tracing simulations was derived from the ResourceCode hindcast at the Bretagne Sud 1 point (47.32°N, 3.55°W) [6]. In Part I, the extremes of this hindcast were analysed using three methods: block maxima of annual significant wave height, a peaks-over-threshold (POT) analysis, and a principal-component-analysis (PCA) environmental contour that preserves the joint distribution of significant wave height and peak period. For a given return period, the PCA contour is a curve in $(H_{\text{s}}, T_{\text{p}})$ space on which all sea states share the same exceedance probability.

For each return period (1 and 50 years), the PCA contour was computed from the full sea-state time series and the point with the *largest* H_{s} on that contour was selected, together with its associated T_{p}. This yields the most severe joint combination of height and period compatible with the contour, rather than a typical contour point, and is therefore intentionally conservative: both H_{s} and T_{p} are larger than those from marginal (univariate) extreme-value fits.

Wave direction was taken from the POT analysis of extreme events, which is based on 137 declustered peaks and is therefore more robust than the 27 annual maxima used for block maxima. This analysis yields a mean coming-from direction of $260.4 \pm 12.6^\circ$ (west–south-west), which is adopted for both return periods. Marginal return levels from the block-maxima and POT fits were retained for comparison but not used directly, because the geometric-optics ray model depends only on period and direction, not on H_{s}.

2) Results and Interpretation: The PCA 1-year design state has $H_\mathrm{s} = 12.35\,\mathrm{m}$ and $T_\mathrm{p} = 20.39\,\mathrm{s}$, and the 50-year state has $H_\mathrm{s} = 18.62\,\mathrm{m}$ and $T_\mathrm{p} = 23.79\,\mathrm{s}$, both with mean direction $260.4°$. A simple consistency check using

$$T_\mathrm{check} \approx 12.7\sqrt{\frac{H_\mathrm{s}}{g}}$$

gives $T_\mathrm{check} \approx 14.3\,\mathrm{s}$ and $17.5\,\mathrm{s}$ for the 1-year and 50-year heights, respectively [7]. The PCA periods are therefore substantially longer than these empirical values and than the typical $T_\mathrm{p} \approx 12$–$18\,\mathrm{s}$ from the block-maxima and POT marginals. This is expected: selecting the uppermost point on each environmental contour forces H_s to be large and pairs it with the long-period tail of the observed H_s–T_p correlation. Because T_check is only an approximate formula, the fact that $T_\mathrm{p} > T_\mathrm{check}$ reflects the use of an extreme joint sea state rather than an inconsistency in the data.

Table II: Offshore design conditions at the Bretagne Sud 1 hindcast point used to force the ray-tracing simulations. Directions are “coming-from,” measured clockwise from north.

Case	H_s (m)	T_p (s)	Direction (deg)
1-year	12.35	20.39	260.4 (west–south-west)
50-year	18.62	23.79	260.4 (west–south-west)

These offshore design sea states therefore represent the most severe joint wave conditions expected once in 1 and 50 years, while remaining consistent with the hindcast joint statistics, and form the basis for the subsequent ray-tracing and load calculations.

B. Bathymetry and Model Domain

1) Methodology: The bathymetry for the ray-tracing simulations was taken from the Global Multi-Resolution Topography (GMRT) grid for the Bretagne Sud region [8]. The data set is an ASCII raster with 1225 columns and 703 rows in geographic coordinates (WGS84). Each cell spans $0.0011°$ in latitude and longitude, which corresponds to a horizontal resolution of about $83\,\mathrm{m}$ east–west and $122\,\mathrm{m}$ north-south at this latitude. A local Cartesian system (x, y) in metres was obtained by converting these angular spacings using Earth radius and cosine-of-latitude factors, with the south-west grid corner taken as the origin.

Water depth was defined as positive downwards from the mean sea level, and land cells were masked. No smoothing, clipping, or other modification of the GMRT bathymetry was applied, so the model retains narrow channels, steep banks, and small islands. The Bretagne Sud 1 hindcast point was projected onto this grid to determine the local coordinates and depth at the project site, which are used later when interpreting ray density and estimating local wave heights.

2) Results and Interpretation: Figure 5 shows the model-domain bathymetry and topography, with depth contours and the project site marked. The axes indicate the local x and y coordinates in metres from the south-west corner, and colour indicates depth, with light tones representing shallow areas and dark tones representing deep water.

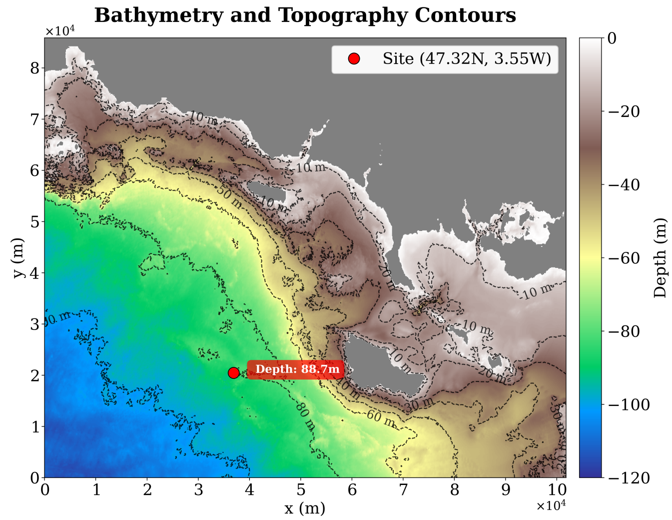

Figure 5: Bathymetry and topography of the Bretagne Sud model domain with depth contours every 10–20 m. The red marker denotes the project site at $(x, y) \approx (3.6 \times 10^4, 2.1 \times 10^4)\,\mathrm{m}$, where the water depth is about $88.7\,\mathrm{m}$.

The bathymetry in figure 5 shows a gently sloping continental shelf that deepens from the inner coastline, where depths are typically less than $20\,\mathrm{m}$, toward the south-west corner, where depths exceed $100\,\mathrm{m}$. Depth contours are broadly aligned with the shoreline and form a diagonal band of 50-$80\,\mathrm{m}$ depths extending from the central western boundary toward the south-eastern headland. Superimposed on this regional slope are shoals, islands, and channels that create an intricate pattern of banks and headlands.

The project site lies on the mid-shelf slope at a depth of approximately $88.7\,\mathrm{m}$, between the deeper outer shelf to the south and the shallower inner shelf and archipelago to the east. To the south-west of the site, the contours bend around a broad underwater ridge that is expected to refract incident west–south-westerly swell toward the coast. This geometry indicates that local wave conditions will be shaped by both large-scale refraction over the shelf and more localised focusing and shadowing associated with the banks and islands.

C. Ray-Tracing Runs and Quality Checks

1) Methodology: The ray-tracing model represents surface gravity waves as rays that propagate over a slowly varying depth field, with local wavenumber and group velocity obtained from the linear dispersion relation [5]. Rays are launched along the western boundary of the GMRT domain, at starting points evenly spaced in y, with headings determined from the offshore design direction in table II. The coming-from direction of $260.4°$ (west–south-west) corresponds to a propagation direction of about $80.4°$ clockwise from north, so the rays travel slightly north of east as they cross the shelf.

Three representative peak periods are used to sample the bathymetry. The PCA contour values $T_\mathrm{ray} = 20.39\,\mathrm{s}$ and $T_\mathrm{ray} = 23.79\,\mathrm{s}$ represent the 1-year and 50-year design storms, and a shorter period $T_\mathrm{ray} = 14\,\mathrm{s}$ provides a comparison typical of more frequent North Atlantic storms from the block-maxima and peaks-over-threshold fits. For each period, two simulations are carried out: a 100-ray visualisation to inspect trajectories and domain coverage, and a 1000-ray run for subsequent energy-flux estimation. The total integration time is $T = 9000\,\mathrm{s}$ (2.5 h). For the $T_{\mathrm{ray}} = 14\,\mathrm{s}$ and 1-year ($T_{\mathrm{ray}} = 20.39\,\mathrm{s}$) cases, $n_t = 20{,}000$ time steps are used, giving $\Delta t = T/n_t \approx 0.45\,\mathrm{s}$ and resolving each period by about 31 and 45 steps, respectively. For the 50-year case ($T_{\mathrm{ray}} = 23.79\,\mathrm{s}$), $n_t = 15{,}000$ so that $\Delta t \approx 0.60\,\mathrm{s}$, corresponding to roughly 40 steps per period. Currents, dissipation, wave breaking, and wind forcing are neglected, so the only processes that modify the rays are refraction and shoaling over the bathymetry.

Treatment of small-scale depth variations in the GMRT grid is an important numerical choice. In the present work, the raw, unsmoothed GMRT bathymetry is used throughout to retain narrow channels, steep banks, and small islands that may influence refraction. A known side effect of this choice is that abrupt depth changes at the single-cell scale can cause individual rays to become trapped in tight loops or to converge onto isolated grid points, a behaviour that is unlikely to represent a physical caustic. A possible improvement, left for future work, would be to construct an alternative bathymetry by applying a modest Gaussian filter with standard deviation of two grid cells ($\sigma \approx 2$, corresponding to an influence radius of order $200\,\mathrm{m}$) and to recompute the rays; such smoothing would be expected to remove the most pathological trajectories while risking some loss of sharp topographic features. Another practical option would be to extend the computational domain slightly further west so that rays are initialised in deeper water rather than in very shallow cells around outer islets. In the discussion of the results below, any anomalous rays are therefore interpreted as artefacts of the unsmoothed grid.

2) Results and Interpretation: The interaction between a shorter-period storm and the shelf is first examined using the 100-ray simulation with $T_{\mathrm{ray}} = 14\,\mathrm{s}$ on the unsmoothed bathymetry. The resulting trajectories are plotted in figure 6.

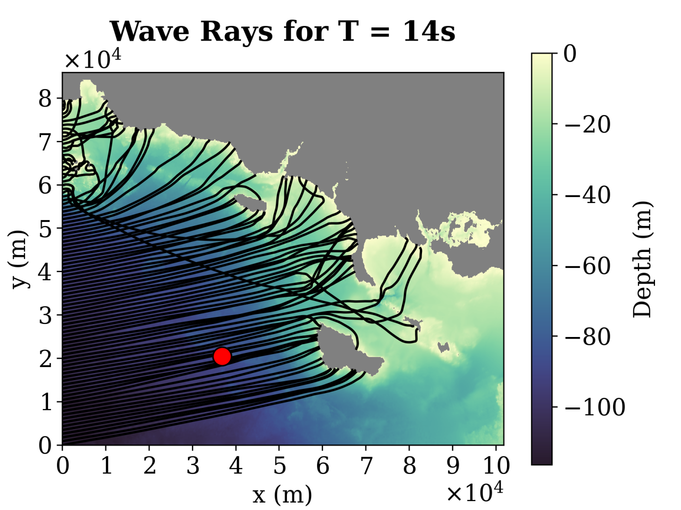

Figure 6: Wave rays for a $T_{\mathrm{ray}} = 14\,\mathrm{s}$ storm. Black curves show ray trajectories over the GMRT bathymetry, and the red marker indicates the project site.

The rays enter along the western boundary between $y \approx 0\,\mathrm{m}$ and $y \approx 8 \times 10^4\,\mathrm{m}$ and cross the shelf in a broad, gently curved fan. Noticeable bending occurs only close to the mid-shelf ridge, and the coastal headlands, and the project site lies within a wide corridor of paths rather than in a sharply focused bundle. For $T_{\mathrm{ray}} = 14\,\mathrm{s}$ the deep-water wavelength is $L_0 \approx 3.1 \times 10^2\,\mathrm{m}$, so launch depths of 80–120 m correspond to $h/L_0 \approx 0.26$–0.39. This intermediate-depth regime implies relatively weak bottom control on the rays, consistent with the mild refraction and the absence of strong focusing at the site in figure 6.

The response of longer-period swell is then investigated using the 1-year design period $T_{\mathrm{ray}} = 20.39\,\mathrm{s}$. The 100-ray trajectories for this case are shown in figure 7.

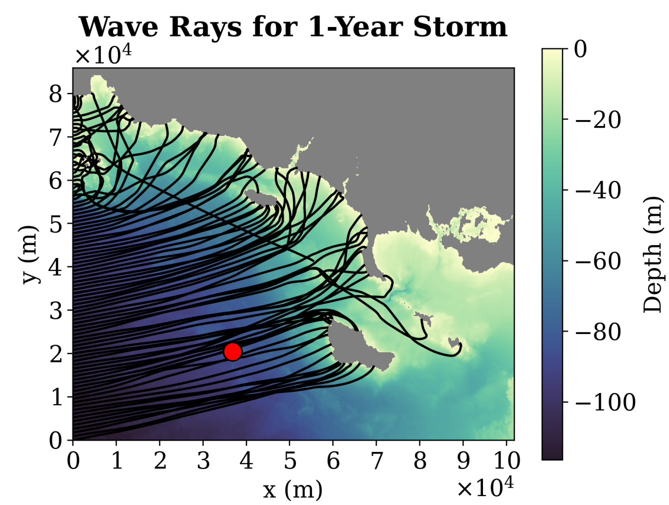

Figure 7: Wave rays for the 1-year storm case with $T_{\mathrm{ray}} = 20.39\,\mathrm{s}$. The red marker denotes the project site.

The rays again sweep across the full domain, but their curvature increases markedly as they cross the $50 - 80\,\mathrm{m}$ depth band, and they wrap more tightly around the shoals and headlands than in figure 6. A dense bundle now passes directly over the project site, which lies near the centre of a converging group of paths. The deep-water wavelength for this period is $L_0 \approx 6.5 \times 10^2\,\mathrm{m}$, so the launch depths correspond to $h/L_0 \approx 0.12$–0.18, indicating that the waves are relatively shallower and hence more sensitive to depth gradients than in the 14 s case. This explains why the mid-shelf ridge plays a stronger role in steering rays toward the site in figure 7.

The 50-year design storm with $T_{\mathrm{ray}} = 23.79\,\mathrm{s}$ is finally considered to see how the rarest, longest-period waves interact with the same bathymetry. The 100-ray visualisation for this case is presented in figure 8.

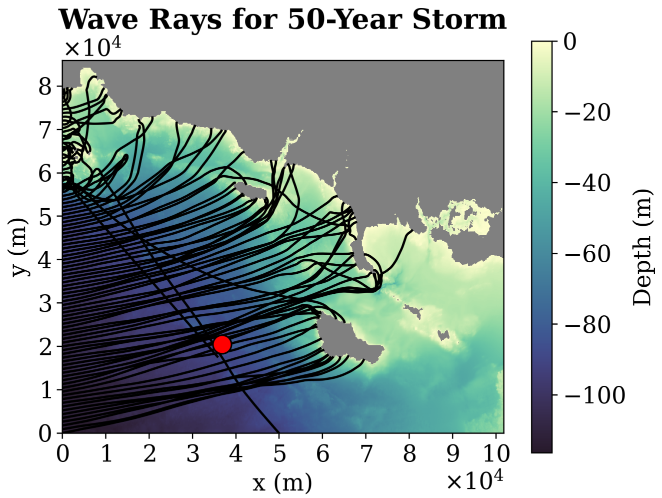

Figure 8: Wave rays for the 50-year storm case with $T_{\mathrm{ray}} = 23.79\,\mathrm{s}$. The project site is marked by the red circle.

The overall structure closely resembles figure 7, but refraction starts further offshore, and the ray bundles wrap more tightly around the shoals south-west of the project site. Several groups of rays converge landward of these banks, forming a narrow focusing band whose flank intersects the site. Two trajectories, however, stand out as clear anomalies: they are initialised in very shallow water close to a small offshore island and then travel away from the coast rather than towards it, in contrast with the neighbouring rays. These anomalous paths are consistent with expectations for rays launched directly on sharp depth gradients in an unsmoothed grid and are therefore interpreted as numerical artefacts of the bathymetry representation rather than physically meaningful caustics. In a refined model, such behaviour could be mitigated either by modest Gaussian smoothing of the bathymetry or by extending the domain westward so that all rays are launched in sufficiently deep, slowly varying water, but these modifications lie outside the scope of the present simulations.

The depth regime at the western boundary provides a compact way to summarise these trends. The deep-water wavelengths for $T_{\mathrm{ray}} = 14\,\mathrm{s}$, $20.39\,\mathrm{s}$, and $23.79\,\mathrm{s}$ are approximately $L_0 \approx 3.1 \times 10^2\,\mathrm{m}$, $6.5 \times 10^2\,\mathrm{m}$, and $8.8 \times 10^2\,\mathrm{m}$, respectively. With launch depths of $h \approx 80$–$120\,\mathrm{m}$, the corresponding depth ratios are $h/L_0 \approx 0.26$–0.39 for 14 s, 0.12–0.18 for the 1-year case, and 0.09–0.14 for the 50-year case. All three are in the intermediate-water regime, but the longer-period waves are relatively shallower and therefore more strongly refracted by depth gradients. The visualisations in figures 6–8, together with the discussion of potential smoothing and domain extensions, show that this increasing sensitivity leads to a persistent focusing corridor that passes over the project site for the 1-year and 50-year design storms, whereas shorter-period storms produce a broader, more weakly focused fan of rays. This provides the geometric basis for interpreting the energy-flux maps and local wave-height amplification at the site in the following subsection.

D. Wave Energy Flux Distribution

1) Methodology: The ray-tracing results were next used to quantify how refraction redistributes wave energy across the shelf and how this redistribution depends on wave period. For each of the three cases, the positions from the 1000-ray runs were mapped onto a regular (x, y) grid with a cell size of order $20\,\mathrm{m}$. The number of ray passages through each cell was counted, giving a ray-density field that is proportional to the depth-integrated energy flux

$$F = E\, c_{\mathrm{g}},$$

where E is the wave energy per unit surface area and c_{g} is the group velocity, under the assumption that energy is conserved along each ray apart from geometric spreading [4]. This field was interpolated back onto the original bathymetry grid and normalised by a reference value along the western boundary, yielding a dimensionless relative flux

$$\frac{E}{E_0} = \frac{F}{F_0},$$

where E_0 and F_0 denote the offshore energy density and flux. Because the ray equations depend only on period and direction, these relative fields can later be combined with different offshore H_{s} values to represent alternative extreme scenarios.

2) Results and Interpretation: The first question is how a shorter-period storm distributes energy over the shelf, and whether the project site is already favoured relative to its surroundings. To investigate this, the ray-density procedure was applied to the 14 s run, and the resulting relative flux field was plotted over the bathymetry.

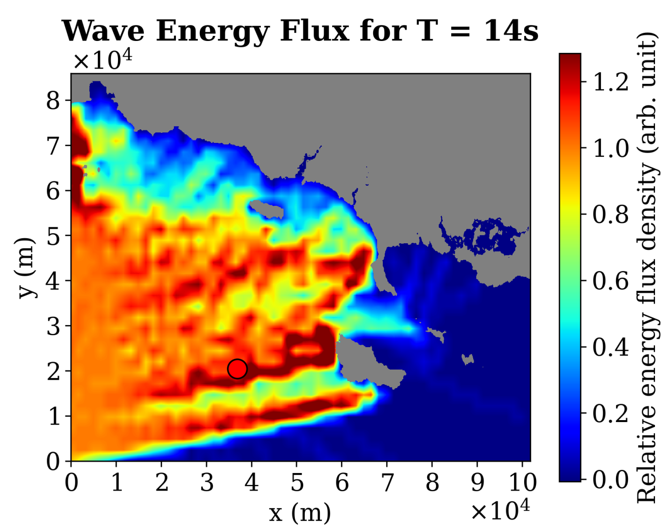

Figure 9: Relative wave energy flux density E/E_0 for a $T_{\mathrm{ray}} = 14\,\mathrm{s}$ storm. The red circle marks the project site.

Figure 9 shows a broad zone of moderately elevated energy extending from the south-western boundary across the mid-shelf, with E/E_0 typically in the range 0.7–1.0. The project site lies near the centre of this band, with E/E_0 at the site around 1.1. Shadowing in the lee of the islands and headlands is present but weak, with E/E_0 rarely dropping below about

0.4. Overall, the 14 s storm produces a relatively diffuse pattern: the site is only slightly focused.

The second question is how this picture changes when the period is increased to the 1-year design value. The same analysis was applied to the 1-year, 1000-ray simulation.

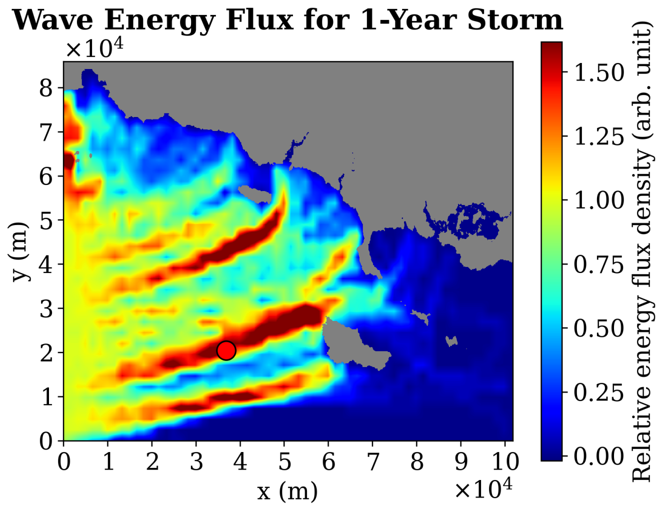

Figure 10: Relative wave energy flux density E/E_0 for the 1-year ray-tracing case with $T_{\mathrm{ray}} = 20.39\,\mathrm{s}$.

In figure 10, the energy field organises into three narrow, elongated bands of high flux that trace the mid-shelf ridge and the banks south-west of the site. Within these caustics, E/E_0 locally reaches about 1.4–1.5, while adjacent embayments and island lees show much lower values, with $E/E_0 \lesssim 0.3$ in some pockets. The project site lies squarely within one of the high-flux bands, and the local E/E_0 is estimated to be around 1.2–1.4. Compared with figure 9, refraction has therefore intensified the spatial contrasts and moved the site from a broadly average location to one of enhanced energy.

The final question is whether the rarer 50-year storm further amplifies this focusing or simply shifts the pattern. The 50-year, 1000-ray simulation was processed in the same way.

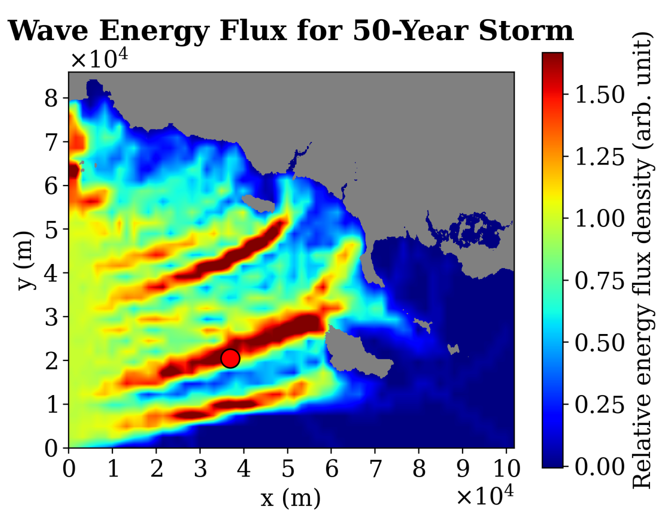

Figure 11: Relative wave energy flux density E/E_0 for the 50-year ray-tracing case with $T_{\mathrm{ray}} = 23.79\,\mathrm{s}$.

Figure 11 shows a structure similar to figure 10, but the high-flux bands are slightly narrower and more intense, with peak values again approaching 1.4–1.5 and broader shadow zones with $E/E_0 \lesssim 0.3$ behind the islands. The project site remains embedded in the central focusing band, with local E/E_0 comparable to, or slightly higher than, that in the 1-year case. The modest shift and sharpening of the caustics are consistent with the stronger refraction observed in figure 8.

Taken together, figures 9–11 demonstrate a clear period dependence: as the peak period increases from 14 s to the 1-year and 50-year design storms, the west–south-westerly swell is progressively focused into narrower bands, and the project site moves from a region of near-average energy into the core of a persistent focusing corridor. This implies that, for a fixed offshore sector, local significant wave heights and associated loads at the site are amplified more strongly under long-period, rare storms than under shorter-period, more frequent events.

E. *Wave Height Estimates at the Project Site*

1) Methodology: The ray-density maps in figures 10 and 11 were used to convert offshore contour conditions into local significant wave heights at the project site. Under linear theory, the depth-integrated energy flux satisfies

$$F = E c_{\mathrm{g}} \propto H_{\mathrm{s}}^2 c_{\mathrm{g}},$$

where E is the wave energy per unit surface area, c_{g} is the group velocity, and H_{s} is the significant wave height. Between the western boundary (depth ≈ 100–$120\,\mathrm{m}$) and the site ($h = 88.7\,\mathrm{m}$), the change in c_{g} for the long periods considered is only a few per cent, so the local height can be approximated from the relative energy factor as

$$\frac{H_{\mathrm{s,loc}}}{H_{\mathrm{s,off}}} \approx \sqrt{\frac{E}{E_0}},$$

where $H_{\mathrm{s,off}}$ is the offshore contour value and E/E_0 is the ray-based energy amplification at the site.

For each return period, E/E_0 was read from the corresponding map at the project location. The 1-year field in figure 10 places the site within a high-flux band with $E/E_0 \approx 1.3$, while the 50-year field in figure 11 shows slightly stronger focusing with $E/E_0 \approx 1.4$. These factors were combined with the offshore contour heights in table II to obtain local $H_{\mathrm{s,loc}}$. Linear dispersion at $h = 88.7\,\mathrm{m}$ was then used to estimate the associated wavelengths for the contour peak periods, providing a consistency check on the use of linear theory.

2) Results and Interpretation: For the 1-year design storm, the offshore contour gives $H_{\mathrm{s,off}} = 12.35\,\mathrm{m}$ and $T_{\mathrm{p}} = 20.39\,\mathrm{s}$. With $E/E_0 \approx 1.3$ at the site,

$$H_{\mathrm{s,loc}} \approx H_{\mathrm{s,off}} \sqrt{\frac{E}{E_0}} \approx 12.35\sqrt{1.3}\,\mathrm{m} \approx 14.1\,\mathrm{m}.$$

Solving the dispersion relation at $h = 88.7\,\mathrm{m}$ for $T_{\mathrm{p}} = 20.39\,\mathrm{s}$ gives a local wavelength $L \approx 5.2 \times 10^2\,\mathrm{m}$, so that $h/L \approx 0.17$ and $H_{\mathrm{s,loc}}/L \approx 0.027$. The waves are therefore in intermediate water with modest steepness, well within the usual linear-theory threshold $H/L \lesssim 0.05$ [9].

For the 50-year storm, the offshore contour values are $H_{\mathrm{s,off}} = 18.62\,\mathrm{m}$ and $T_{\mathrm{p}} = 23.79\,\mathrm{s}$. The site lies in a slightly stronger focusing band with $E/E_0 \approx 1.4$, which yields

$$H_{\mathrm{s,loc}} \approx 18.62\sqrt{1.4}\,\mathrm{m} \approx 22.0\,\mathrm{m}.$$

At the same depth, the dispersion relation for $T_{\mathrm{p}} = 23.79\,\mathrm{s}$ gives $L \approx 6.3 \times 10^2\,\mathrm{m}$, so $h/L \approx 0.14$ and $H_{\mathrm{s,loc}}/L \approx 0.035$. This again corresponds to intermediate water and moderate steepness.

These estimates show that bathymetric focusing at the project site increases the local significant wave height by roughly $15\,\%$ for the 1-year storm and about $20\,\%$ for the 50-year storm relative to the offshore contour heights. At the same time, the local wave parameters remain comfortably within the regime where linear ray tracing and linear kinematics are appropriate, supporting their use in the subsequent monopile load calculations.

F. Summary of Design Conditions at the Project Site

1) Methodology: To provide a compact set of inputs for the wave-load analysis, the offshore environmental contour states were combined with the ray-tracing amplification factors to define local design sea states at the project site. The location was fixed at the bathymetric grid point corresponding to $(47.32°\mathrm{N}, 3.55°\mathrm{W})$ with water depth $h = 88.7\,\mathrm{m}$. For each return period, the local significant wave height $H_{\mathrm{s,loc}}$ was taken from the ray-based estimates above, the peak period T_{p} from the PCA environmental contour, and the mean direction from the POT directional analysis, giving a consistent set of site-specific design parameters [1].

2) Results and Interpretation: Table III summarises the resulting local design sea states after accounting for bathymetric refraction and focusing.

Table III: Local design sea states at the project site obtained by combining offshore environmental-contour conditions with ray-tracing-derived energy amplification. Directions are "coming-from," measured clockwise from north.

Case	h (m)	$H_{\mathrm{s,loc}}$ (m)	T_{p} (s)	Direction (deg)	Sector
1-year	88.7	14.1	20.39	260.4	WSW
50-year	88.7	22.0	23.79	260.4	WSW

Both design states correspond to long-period swell in intermediate water, with local heights amplified relative to the offshore contours by refraction over the mid-shelf ridge and surrounding banks. The 50-year storm is characterised by substantially larger $H_{\mathrm{s,loc}}$ and longer T_{p} than the 1-year case, as well as stronger focusing at the site. These local sea states provide a physically consistent bridge between the hindcast-based extreme-value analysis and future load calculations, ensuring that the loads are evaluated under wave conditions that reflect both offshore extremes and the spatial transformation of the wave field over the complex bathymetry.

IV. Wave Loads on a Vertical Cylinder

The purpose of this section is to connect laboratory measurements of wave forces on a vertical cylinder to the corresponding loads on a scaled monopile in the ocean. The analysis combines linear wave theory, dimensional analysis, and a linear diffraction estimate of the inline force. The experimental arrangement in the flume is shown in figure 12.

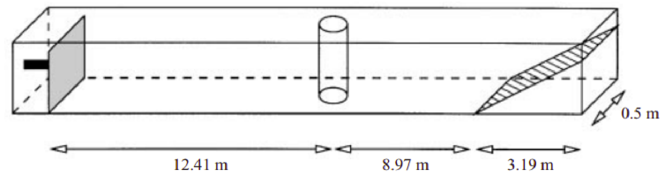

Figure 12: Wave tank for the case when the small cylinder is placed 12.41 m from the wave maker [10].

Waves are generated by a wavemaker on the left and absorbed on the right, with a vertical circular cylinder mounted on the flume centreline. The still-water depth in the constant-depth section is $h = 0.60\,\mathrm{m}$ and the tank width is $0.50\,\mathrm{m}$. Measured along the tank centreline, the cylinder centre is $12.41\,\mathrm{m}$ from the wavemaker, followed by a further $8.97\,\mathrm{m}$ of constant depth before the start of the sloping beach. The beach has a horizontal length of $3.19\,\mathrm{m}$ and reduces the depth from $0.60\,\mathrm{m}$ to zero. Two cylinders are tested, with radii $R = 0.03\,\mathrm{m}$ and $R = 0.04\,\mathrm{m}$ respectively.

A. Wavelengths and Water Depth Regime

The objective here is to determine the wavelengths corresponding to all model test frequencies and to classify each case as deep, intermediate, or shallow water relative to the tank depth. The imposed regular waves have cyclic frequencies between about $1\,\mathrm{Hz}$ and $1.6\,\mathrm{Hz}$, and the water depth in the constant-depth region is uniform with $h = 0.60\,\mathrm{m}$. Linear (Airy) wave theory is adopted, which is appropriate for the modest wave steepnesses and for the regular monochromatic forcing in the flume.

For a wave with cyclic frequency f, the angular frequency is $\omega = 2\pi f$, and the dispersion relation in constant depth h is

$$\omega^2 = gk\tanh(kh),$$

where $g = 9.81\,\mathrm{m\,s^{-2}}$ is gravitational acceleration, k is the wavenumber, and kh is a measure of relative depth [4]. For each experimental frequency, the dispersion relation is solved numerically for $k(f)$ at $h = 0.60\,\mathrm{m}$, and the corresponding wavelength $\lambda(f)$ and depth ratio h/λ are then obtained from

$$\lambda(f) = \frac{2\pi}{k(f)}, \qquad \frac{h}{\lambda(f)} = \frac{h\,k(f)}{2\pi}.$$

For the cylinder with radius $R = 0.03\,\mathrm{m}$ the imposed frequencies are

$$f = 1.171,\ 1.300,\ 1.425,\ 1.615\,\mathrm{Hz}.$$

Solving the dispersion relation gives wavelengths

$$\lambda \approx 1.14,\ 0.923,\ 0.769,\ 0.599\,\mathrm{m},$$

which correspond to depth ratios

$$\frac{h}{\lambda} \approx 0.53,\ 0.65,\ 0.78,\ 1.00.$$

All four cases satisfy $h/\lambda \gtrsim 0.5$, which places them in the deep-water regime.

For the cylinder with radius $R = 0.04\,\mathrm{m}$ the imposed frequencies are

$$f = 1.009,\ 1.123,\ 1.399,\ 1.532\,\mathrm{Hz},$$

which yield wavelengths

$$\lambda \approx 1.51,\ 1.23,\ 0.798,\ 0.665\,\mathrm{m},$$

and depth ratios

$$\frac{h}{\lambda} \approx 0.40,\ 0.49,\ 0.75,\ 0.90.$$

The two lowest-frequency cases have $h/\lambda < 0.5$ and are therefore in intermediate depth, while the two higher-frequency cases satisfy $h/\lambda \gtrsim 0.5$ and are classified as deep-water waves [4].

Therefore, all model tests are conducted in deep or intermediate water, with most conditions falling within the deep-water regime. No test case is in shallow water, so shallow-water approximations are not used later.

B. Non-Dimensional Parameter kR

The objective of this subsection is to quantify how small the cylinder is relative to the wavelength and to identify whether the flow regime corresponds to a long-wave diffraction regime, characterised by a slender-cylinder, inertia-dominated response or to a regime where short-wave diffraction becomes important. This is achieved using the non-dimensional size parameter kR, which is directly obtained from the wavenumbers calculated in section IV-A and the two model radii.

The usual slenderness parameter in wave–structure interaction is the diameter-to-wavelength ratio D/λ, which appears explicitly in the MacCamy–Fuchs diffraction solution through the added-mass function $C_{\mathrm{M}}(D/\lambda)$ [2]. For small D/λ, the MacCamy–Fuchs inertia coefficient tends to the long-wave value $C_M \approx 2$, so that diffraction mainly acts to fine-tune the inertia term rather than to change the overall force structure.

The parameter kR is directly proportional to D/λ and therefore provides an equivalent measure of slenderness. Substituting $k = 2\pi/\lambda$ and $D = 2R$ gives

$$kR = \frac{2\pi}{\lambda}R = \pi\,\frac{D}{\lambda},$$

so that

$$\frac{D}{\lambda} = \frac{kR}{\pi}.$$

The condition that the cylinder is "much smaller than the wavelength" in the Morison sense, therefore, corresponds to $D/\lambda \ll 1$, or equivalently $kR \ll \pi$. In practical engineering use, Morison-type and long-wave diffraction formulations are considered appropriate while $D/\lambda \lesssim 0.2$–0.3, which corresponds to $kR \lesssim 0.6$–1.0 [11]. For D/λ of order unity, the structure size becomes comparable to the wavelength, the reflected and scattered wave fields are strong, and full short-wave diffraction theory is required.

For the small cylinder with radius $R = 0.03\,\mathrm{m}$, the wavenumbers in section IV-A give

$$kR \approx 0.166,\ 0.204,\ 0.245,\ 0.315$$

for $f = 1.171,\ 1.300,\ 1.425$, and $1.615\,\mathrm{Hz}$, respectively. These values correspond to diameter-to-wavelength ratios

$$\frac{D}{\lambda} = \frac{kR}{\pi} \approx 0.053,\ 0.065,\ 0.078,\ 0.10,$$

so the cylinder diameter is between about $5\,\%$ and $10\,\%$ of the wavelength. This lies well inside the slender-cylinder regime $D/\lambda \ll 1$, and MacCamy–Fuchs theory would predict an inertia coefficient very close to the long-wave limit $C_{\mathrm{M}} \approx 2$ for these conditions.

For the larger cylinder with radius $R = 0.04\,\mathrm{m}$, the corresponding non-dimensional sizes are

$$kR \approx 0.166,\ 0.204,\ 0.315,\ 0.378$$

for $f = 1.009,\ 1.123,\ 1.399$, and $1.532\,\mathrm{Hz}$, respectively. These translate to

$$\frac{D}{\lambda} = \frac{kR}{\pi} \approx 0.053,\ 0.065,\ 0.10,\ 0.12,$$

so even in the most extreme case, the diameter is only about $12\,\%$ of the wavelength. This is still well below the range $D/\lambda \sim O(1)$ where diffraction dominates, and it is also below the $D/\lambda \lesssim 0.2$ threshold commonly taken as the upper limit of reliable Morison-type modelling.

Consequently, in all eight model-scale test cases, the cylinder is small compared with the wavelength, with $kR \lesssim 0.38$ and $D/\lambda \lesssim 0.12$, so the flow regime corresponds to a slender, inertia-dominated response with relatively weak diffraction. Diffraction effects are present and are accounted for later using MacCamy–Fuchs theory through an effective inertia coefficient $C_{\mathrm{M}}(D/\lambda)$, but the present kR values justify interpreting the laboratory loads primarily within a Morison or long-wave diffraction framework rather than invoking full short-wave diffraction theory.

C. Geometric Similarity and Choice of Model Cylinder

It is helpful to determine which of the two model cylinders is dynamically similar to a "full-scale" prototype cylinder of radius $R_{\mathrm{p}} = 2\,\mathrm{m}$ in water depth $h_{\mathrm{p}} = 40\,\mathrm{m}$. The key idea from dimensional analysis is that model and prototype should share the same dominant non-dimensional parameters: geometrically, the ratios of water depth to structural size (for example h/R), and dynamically the Froude number, so that the role of gravity and the free surface is reproduced. Viscous effects are neglected, so Reynolds similarity is not enforced, which is consistent with wave–structure interaction at large Reynolds numbers in water.

The primary geometric similarity requirement in this configuration is that the depth-to-radius ratio should be the same at model and full scale. For the full-scale prototype cylinder,

$$\frac{h_{\mathrm{p}}}{R_{\mathrm{p}}} = \frac{40}{2} = 20.$$

For the small model cylinder with radius $R_{\mathrm{m}} = 0.03\,\mathrm{m}$ in water depth $h_{\mathrm{m}} = 0.60\,\mathrm{m}$, the corresponding model ratio is

$$\frac{h_{\mathrm{m}}}{R_{\mathrm{m}}} = \frac{0.60}{0.03} = 20,$$

which matches the prototype value exactly. Here, subscripts m and p denote model and prototype quantities, respectively. For the larger cylinder with radius $R_\mathrm{m} = 0.04\,\mathrm{m}$, the ratio is 15, which does not reproduce the full-scale geometry, because the model cylinder is comparatively larger relative to the water depth than the prototype monopile.

Froude scaling is adopted so that prototype and model share the same Froude number,

$$\mathrm{Fr} = \frac{U}{\sqrt{gL_\mathrm{c}}},$$

where U is a characteristic wave-induced horizontal velocity and L_c is a characteristic length, taken here as the water depth, so that $L_\mathrm{c} = h$ [12]. Geometric similarity fixes a single length scale factor

$$\lambda_\ell = \frac{h_\mathrm{p}}{h_\mathrm{m}} = \frac{40}{0.60} \approx 66.7,$$

and all other lengths, including the cylinder radius, scale with the same factor. Imposing $\mathrm{Fr}_\mathrm{p} = \mathrm{Fr}_\mathrm{m}$ then requires that the characteristic velocities and periods satisfy

$$\frac{U_\mathrm{p}}{U_\mathrm{m}} = \sqrt{\lambda_\ell}, \qquad \frac{T_\mathrm{p}}{T_\mathrm{m}} = \sqrt{\lambda_\ell},$$

which is the time scaling used in the following subsection (IV-D) when converting the model wave period to the prototype period. This shows that the 3 cm model cylinder is consistent with a single Froude scale factor between depth and radius, whereas the 4 cm cylinder would imply a different scale for the structure than for the water depth and would therefore break geometric similarity.

In conclusion, the experimental results obtained with the $R = 0.03\,\mathrm{m}$ cylinder can be interpreted as a dynamically similar model of a $R_\mathrm{p} = 2\,\mathrm{m}$ monopile in $h_\mathrm{p} = 40\,\mathrm{m}$ water depth under Froude scaling with length factor $\lambda_\ell \approx 66.7$. The $R = 0.04\,\mathrm{m}$ cylinder does not satisfy this geometric requirement for the chosen prototype and so is not used for direct scaling to the specified full-scale prototype monopile.

D. Prototype Wave Period and Wavelength

The objective of this subsection is to determine the wave period and wavelength at full scale corresponding to the regular wave shown in figure 13, so that the laboratory force measurements can be related to a specific prototype sea state.

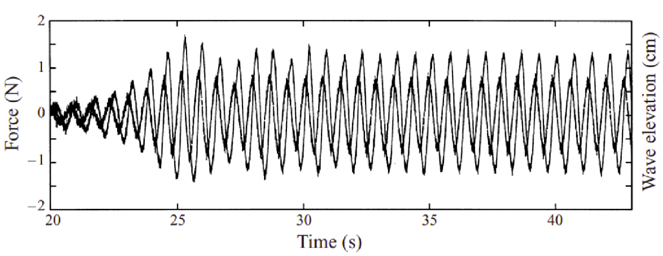

Figure 13: Time history of the wave elevation and the force on the small cylinder 12.41 m from the wave maker. The time window used for the force measurements is from 35.088 s to 42.105 s, *i.e.*, 10 wave periods. The curve with the greater amplitude represents the wave elevation. Here $f = 1.425$ Hz and $k_\mathrm{m}a_\mathrm{m} = 0.10$ [10].

Figure 13 shows concurrent time series of inline force and free-surface elevation at the cylinder location, together with the time window used for the force analysis. The imposed model wave frequency is $f_\mathrm{m} = 1.425\,\mathrm{Hz}$, so the corresponding model wave period is

$$T_\mathrm{m} = \frac{1}{f_\mathrm{m}} \approx \frac{1}{1.425}\,\mathrm{s} \approx 0.702\,\mathrm{s},$$

which is consistent with the 10-wave-period interval between $t = 35.088\,\mathrm{s}$ and $t = 42.105\,\mathrm{s}$ in figure 13.

Using the Froude time scaling from subsection IV-C, with $\lambda_\ell \approx 66.7$, the prototype wave period is

$$T_\mathrm{p} = \sqrt{\lambda_\ell}\,T_\mathrm{m} \approx \sqrt{66.7} \times 0.702\,\mathrm{s} \approx 5.73\,\mathrm{s}.$$

To obtain the prototype wavelength, the dispersion relation is solved at the prototype depth $h_\mathrm{p} = 40\,\mathrm{m}$ using this period. The prototype angular frequency is

$$\omega_\mathrm{p} = \frac{2\pi}{T_\mathrm{p}} \approx \frac{2\pi}{5.73}\,\mathrm{s}^{-1} \approx 1.10\,\mathrm{s}^{-1},$$

and the linear dispersion relation is solved numerically for the prototype wavenumber k_p. This gives

$$k_\mathrm{p} \approx 0.123\,\mathrm{m}^{-1}, \qquad \lambda_\mathrm{p} = \frac{2\pi}{k_\mathrm{p}} \approx 51.3\,\mathrm{m},$$

and the corresponding depth ratio

$$\frac{h_\mathrm{p}}{\lambda_\mathrm{p}} \approx \frac{40}{51.3} \approx 0.78,$$

which lies in the deep-water regime, consistent with the model-scale classification.

Therefore, the $f_\mathrm{m} = 1.425\,\mathrm{Hz}$ model wave corresponds, under Froude scaling, to a prototype regular wave with period $T_\mathrm{p} \approx 5.7\,\mathrm{s}$ and wavelength $\lambda_\mathrm{p} \approx 51\,\mathrm{m}$ in $40\,\mathrm{m}$ water depth.

E. Prototype Wave Amplitude

The objective of this subsection is to determine the wave amplitude at full scale, corresponding to the regular wave in figure 13. The non-dimensional wave steepness is $k_\mathrm{m}a_\mathrm{m} = 0.10$, where a_m is the model wave amplitude and k_m is the model wavenumber for the case with $f_\mathrm{m} = 1.425\,\mathrm{Hz}$ [10]. From section IV-A, the corresponding model wavenumber is $k_\mathrm{m} \approx 8.17\,\mathrm{m}^{-1}$.

The model amplitude is obtained directly from the steepness definition,

$$k_\mathrm{m}a_\mathrm{m} = 0.10 \quad \Rightarrow \quad a_\mathrm{m} = \frac{0.10}{k_\mathrm{m}} \approx \frac{0.10}{8.17}\,\mathrm{m} \approx 0.0122\,\mathrm{m},$$

so the associated model wave height is

$$H_\mathrm{m} = 2a_\mathrm{m} \approx 0.0245\,\mathrm{m}.$$

With the same length scale $\lambda_\ell \approx 66.7$, the prototype amplitude is

$$a_\mathrm{p} = \lambda_\ell a_\mathrm{m} \approx 66.7 \times 0.0122\,\mathrm{m} \approx 0.816\,\mathrm{m},$$

and the corresponding prototype wave height is

$$H_\mathrm{p} = 2a_\mathrm{p} \approx 1.63\,\mathrm{m}.$$

To summarise, the regular wave with steepness $k_\mathrm{m} a_\mathrm{m} = 0.10$ at $f_\mathrm{m} = 1.425\,\mathrm{Hz}$ corresponds, under Froude scaling, to a prototype regular wave of amplitude $a_\mathrm{p} \approx 0.82\,\mathrm{m}$ and height $H_\mathrm{p} \approx 1.6\,\mathrm{m}$ at the offshore monopile site.

F. Prototype Horizontal Force from Scaled Experiment

The objective of this subsection is to estimate the prototype horizontal force for the conditions derived in sections IV-D and IV-E using only the measured laboratory forces and Froude scaling. The key assumption is that the inline load on the slender vertical cylinder is dominated by inertia rather than drag, which is consistent with the small diameter-to-wavelength ratio D/λ for these tests and with the long-wave diffraction regime identified in section IV-A. Viscous effects are neglected, so Reynolds similarity is not enforced.

The inertia contribution to the inline wave load on a slender vertical cylinder can be expressed in Morison form as

$$F(t) \sim \rho C_\mathrm{M} V_\mathrm{c} \frac{\mathrm{d}u}{\mathrm{d}t},$$

where ρ is the water density, C_M is an effective inertia coefficient, V_c is a characteristic volume proportional to $R^2 h$, and u is a characteristic horizontal water particle velocity [13]. Under geometric similarity, all linear dimensions (for example R and h) scale with the same length factor λ_ℓ, so the characteristic volume scales as

$$V_\mathrm{c,p} = \lambda_\ell^3 V_\mathrm{c,m}.$$

Froude similarity implies that the wave amplitude scales with length, $a_\mathrm{p}/a_\mathrm{m} = \lambda_\ell$, and that the period scales as $T_\mathrm{p}/T_\mathrm{m} = \sqrt{\lambda_\ell}$, so the angular frequency scales as $\omega_\mathrm{p}/\omega_\mathrm{m} = 1/\sqrt{\lambda_\ell}$. The characteristic horizontal acceleration scales as

$$\left|\frac{\partial u_\mathrm{p}}{\partial t}\right| \sim a_\mathrm{p}\omega_\mathrm{p}^2 = \lambda_\ell a_\mathrm{m} \left(\frac{\omega_\mathrm{m}}{\sqrt{\lambda_\ell}}\right)^2 = a_\mathrm{m}\omega_\mathrm{m}^2 \sim \left|\frac{\partial u_\mathrm{m}}{\partial t}\right|,$$

so the acceleration scale factor is unity. With ρ and C_M taken as the same in model and prototype, the inertia force therefore scales purely with the characteristic volume, such that

$$\frac{F_\mathrm{p}}{F_\mathrm{m}} = \lambda_\ell^3.$$

For the present configuration the length scale factor is $\lambda_\ell \approx 66.7$, giving a force scale factor

$$\lambda_\ell^3 \approx (66.7)^3 \approx 2.97 \times 10^5.$$

If the peak horizontal force amplitude measured in the laboratory for the $R = 0.03\,\mathrm{m}$ cylinder at $f_\mathrm{m} = 1.425\,\mathrm{Hz}$ is denoted by F_m^*, the corresponding prototype peak inline force amplitude is

$$F_\mathrm{p}^* \approx \lambda_\ell^3 F_\mathrm{m}^* \approx 2.97 \times 10^5 F_\mathrm{m}^*.$$

Reading the experimental time series in figure 13 gives a representative model-scale peak of about $F_\mathrm{m}^* \approx 1.40\,\mathrm{N}$ for this case, which yields a prototype peak force

$$F_\mathrm{p}^* \approx 2.97 \times 10^5 \times 1.40\,\mathrm{N} \approx 4.16 \times 10^5\,\mathrm{N}.$$

Consequently, for the regular wave corresponding to $f_\mathrm{m} = 1.425\,\mathrm{Hz}$, the scaled experiment predicts a prototype peak horizontal force on the $R_\mathrm{p} = 2\,\mathrm{m}$ monopile of order $4 \times 10^5\,\mathrm{N}$. This estimate relies on the inertia-dominated assumption and the Froude scaling factor λ_ℓ^3 and is shown in the next subsection to be consistent with the MacCamy–Fuchs diffraction prediction for the same prototype sea state.

G. Prototype Horizontal Force from MacCamy–Fuchs Theory

This subsection uses linear diffraction theory in the form of the MacCamy–Fuchs solution to predict the inline force on the prototype monopile. In the long-wave regime relevant here, MacCamy–Fuchs theory can be written as an effective inertia coefficient applied to the Morison-type inertia term, which modifies the Froude–Krylov load to account for diffraction while retaining linear wave kinematics.

For the prototype bottom-mounted cylinder of radius R_p in water depth h_p defined in subsection IV-C, subjected to the regular wave characterised in subsections IV-D and IV-E, the inline inertia force in linear theory, integrated over the cylinder draft, can be written as

$$F_\mathrm{MF}(t) = F_0 \cos(\omega_\mathrm{p} t + \phi)\,,$$

with amplitude

$$F_0 = \rho C_\mathrm{M} \pi R_\mathrm{p}^2 \frac{\omega_\mathrm{p}^2 a_\mathrm{p}}{k_\mathrm{p}},$$

where ρ is the water density, C_M is an effective inertia coefficient, $\omega_\mathrm{p} = 2\pi/T_\mathrm{p}$ is the prototype angular frequency, and k_p is the prototype wavenumber. This expression follows from inserting the linear horizontal acceleration profile from Airy theory into the inertia term, integrating from the seabed $z = -h_\mathrm{p}$ to the still water level $z = 0$, and using

$$\int_{-h_\mathrm{p}}^{0} \cosh[k_\mathrm{p}(z + h_\mathrm{p})]\,\mathrm{d}z = \frac{1}{k_\mathrm{p}} \sinh(k_\mathrm{p} h_\mathrm{p})\,,$$

together with the dispersion relation $\omega_\mathrm{p}^2 = g k_\mathrm{p} \tanh(k_\mathrm{p} h_\mathrm{p})$ to replace $g \tanh(k_\mathrm{p} h_\mathrm{p})$ by $\omega_\mathrm{p}^2/k_\mathrm{p}$.

As found in subsection IV-D, the full-scale prototype is in the deep water regime. The slenderness parameter for the prototype cylinder is $k_\mathrm{p} R_\mathrm{p} \approx 0.245$, which lies in the long-wave diffraction regime where MacCamy–Fuchs theory gives an effective inertia coefficient close to

$$C_\mathrm{M} \approx 2,$$

so that the inertia load is about twice the Froude–Krylov force on the displaced water column [2]. Taking $\rho \approx 1025\,\mathrm{kg\,m^{-3}}$, $R_\mathrm{p} = 2\,\mathrm{m}$, $a_\mathrm{p} \approx 0.816\,\mathrm{m}$, $\omega_\mathrm{p} \approx 1.10\,\mathrm{s^{-1}}$, and $k_\mathrm{p} \approx 0.123\,\mathrm{m^{-1}}$, the force amplitude evaluates to

$$F_0 \approx 1025 \times 2 \times \pi \times (2)^2 \times \frac{(1.10)^2 \times 0.816}{0.123}\,\mathrm{N} \approx 2.1 \times 10^5\,\mathrm{N}.$$

Thus, for the regular wave corresponding to the $f_\mathrm{m} = 1.425\,\mathrm{Hz}$ model case, MacCamy–Fuchs diffraction theory predicts a peak inline horizontal force on the $R_\mathrm{p} = 2\,\mathrm{m}$ monopile of order $2 \times 10^5\,\mathrm{N}$, which can be compared directly with the value obtained by scaling the measured laboratory forces in the preceding subsection to assess the consistency between experiment and linear diffraction theory.

V. CONCLUSION

This report has extended the hindcast-based climate and extreme-value analysis of Part I into a spatially and structurally resolved description of wave conditions and loads at the Bretagne Sud 1 site. The aim was to move beyond pointwise offshore statistics by combining satellite observations, geometric-optics ray tracing over realistic bathymetry, and laboratory wave-tank data, so as to construct a physically consistent path from large-scale sea states to local monopile-scale forces.

Sentinel-2 imagery on 25 March 2020 provided an independent snapshot of the regional wave field under moderate swell conditions. Long-crested waves arriving from the west–north–west exhibited clear refraction and diffraction around Belle-Ile-en-Mer, confirming strong bathymetric control in the same sector that influences Bretagne Sud 1. Quantitative measurements yielded an observed wavelength $L_{\mathrm{obs}} \approx 238 \pm 45\,\mathrm{m}$ and coming-from direction $\theta_{\mathrm{obs}} \approx 293.1 \pm 3.0°$, consistent with the ResourceCode-based theoretical wavelength $L_{\mathrm{theor}} \approx 272 \pm 30\,\mathrm{m}$ and mean direction $\theta_{\mathrm{RC}} = 286.7 \pm 2.2°$. This agreement supports the use of hindcast peak periods and directions as physically realistic input to the subsequent ray-tracing analysis.

Ray-tracing simulations then showed how PCA-based 1- and 50-year west-southwesterly storms are transformed as they cross the GMRT bathymetry. As the peak period increases, refraction focuses rays into narrow caustics that track the mid-shelf ridge and neighbouring banks, while shadow zones develop in the lee of islands and headlands. The project site lies inside a persistent focusing corridor for $T_{\mathrm{p}} = 20.39\,\mathrm{s}$ and $23.79\,\mathrm{s}$, with relative energy factors $E/E_0 \approx 1.3$ and 1.4, respectively. When combined with offshore contour heights, this focusing leads to local significant wave heights $H_{\mathrm{s,loc}} \approx 14.1\,\mathrm{m}$ for the 1-year storm and $H_{\mathrm{s,loc}} \approx 22.0\,\mathrm{m}$ for the 50-year storm. Both states correspond to intermediate water with moderate steepness, remaining within the range of linear theory while illustrating that long-period extremes are systematically amplified at the turbine site by bathymetry.

The laboratory and theoretical load analysis connected these transformed sea states to structural response at the monopile scale. A $R = 0.03\,\mathrm{m}$ cylinder in $0.60\,\mathrm{m}$ depth was shown to be geometrically and dynamically similar, under Froude scaling with $\lambda_\ell \approx 66.7$, to a $R_{\mathrm{p}} = 2\,\mathrm{m}$ monopile in $40\,\mathrm{m}$ water depth. For the selected test at $f_{\mathrm{m}} = 1.425\,\mathrm{Hz}$, the corresponding prototype regular wave has period $T_{\mathrm{p}} \approx 5.7\,\mathrm{s}$, wavelength $\lambda_{\mathrm{p}} \approx 51\,\mathrm{m}$, and height $H_{\mathrm{p}} \approx 1.6\,\mathrm{m}$, with $D/\lambda \lesssim 0.1$ and $k_{\mathrm{p}} R_{\mathrm{p}} \approx 0.245$. These parameters place the monopile in a long-wave, inertia-dominated diffraction regime where Morison-type formulations augmented by MacCamy–Fuchs theory are appropriate [2], [11]. Under these conditions, MacCamy–Fuchs predicts a peak inline force of order $2 \times 10^5\,\mathrm{N}$, comparable to the value obtained by scaling the measured laboratory forces with the cubic Froude factor λ_ℓ^3. It should be noted that this case corresponds to a moderate prototype sea state rather than a 1- or 50-year extreme; extending the load estimates to the extreme design sea states would require combining the Froude scaling with a representative irregular-sea load model.

Taken together, the results show that a coherent chain can be established from hindcast extremes to local structural loads at Bretagne Sud 1. The offshore environmental contour supplies statistically robust design sea states; satellite imagery confirms that these conditions are realised in the field and reveals the importance of refraction and diffraction; ray tracing quantifies how long-period storms are focused at the site and yields amplified local design heights; and scaled laboratory tests, interpreted through long-wave diffraction theory, translate these sea states into monopile-scale forces. For floating wind design, this chain clarifies that local wave loading is controlled jointly by offshore climate, bathymetric transformation, and structural scale, rather than by deep-water statistics alone.

The methodology developed here suggests several targeted extensions that follow directly from the present analysis. On the wave side, repeating the ray-tracing simulations on a modestly smoothed GMRT bathymetry and on a slightly extended western domain would help separate physically meaningful caustics from artefacts of grid roughness and boundary placement. Applying the same wavelength- and direction-extraction procedure to additional Sentinel-2 overpasses would test how robust the focusing corridor is across different swell events. It would also be valuable to compare design sea states derived from nearby ResourceCode hindcast nodes with the spatial focusing and shadowing patterns predicted by the ray-tracing model. On the structural side, exploiting the full set of laboratory test cases to cover a wider range of kR would allow a more systematic comparison between Froude-scaled forces and MacCamy–Fuchs predictions. These extensions would further strengthen the link between offshore climate statistics, site-specific wave transformation, and monopile-scale loads at Bretagne Sud 1.

DECLARATION OF AI USE

ChatGPT (OpenAI) was used exclusively to refine the manuscript's language, formatting, and structure. All scientific interpretations, analyses, and conclusions remain the sole responsibility of the author.

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REFERENCES

[1] L. Yue Ting Shing, "Wind turbine project: Part i – wave dynamics at bretagne sud," course project report, École Polytechnique, Palaiseau, France, 2025. Sea States, Wave Propagation, and Ocean Wave Energy.

[2] R. C. MacCamy and R. A. Fuchs, "Wave forces on piles: A diffraction theory," Technical Memorandum 69, U.S. Beach Erosion Board, U.S. Army Corps of Engineers, Washington, D.C., 1954. 17 pp.

[3] Copernicus Data Space Ecosystem, "Copernicus data space ecosystem browser." https://browser.dataspace.copernicus.eu/, 2023. Accessed: 27 November 2025.

[4] J. Harris, C. Peyrard, and M. Yates, "Sea states, wave propagation, and ocean wave energy: Class notes, 2025–2026." Course notes for MEC53456EP *Sea States, Wave Propagation and Ocean Wave Energy*, LHSV, École nationale des ponts et chaussées and EDF R&D, 2025. Document: SSWPOWE_Class_Notes.pdf.

[5] T. Halsne, K. H. Christensen, G. Hope, and Ø. Breivik, "Ocean wave tracing v.1: a numerical solver of the wave ray equations for ocean waves on variable currents at arbitrary depths," *Geoscientific Model Development*, vol. 16, no. 22, pp. 6515–6530, 2023.

[6] M. Accensi, M. Alday, and C. Maisondieu, "RESOURCECODE: Wave hindcast database user manual," Technical report Issue 1.3, Ifremer, Mar. 2022.

[7] J. Harris, C. Peyrard, and M. Yates, "Ray tracing exercise (Part 1/2), 2024–2025." Course exercise sheet for Sea States, Wave Propagation and Ocean Wave Energy (SSWPOWE), 2025. Document: SSWPOWE_raytracing_2025-1.pdf.

[8] "GMRT MapTool." https://www.gmrt.org/GMRTMapTool/. Global Multi-Resolution Topography (GMRT) synthesis web application, accessed 27 November 2025.

[9] I. R. Young, *Wind Generated Ocean Waves*, vol. 2 of *Elsevier Ocean Engineering Series*. Amsterdam: Elsevier, 1999.

[10] J. Harris, C. Peyrard, and M. Yates, "Wave loads exercise." Course handout, *Sea States, Wave Propagation, and Ocean Wave Energy (SSWPOWE)*, 2024. Wave load estimation exercise sheet.

[11] H. Santo, P. H. Taylor, C. H. K. Williamson, and Y. S. Choo, "The relative-velocity version of the Morison equation for obstacle arrays in combined steady, low and high frequency motion," *Journal of Fluid Mechanics*, vol. 842, pp. 188–214, 2018.

[12] V. Heller, "Scale effects in physical hydraulic engineering models," *Journal of Hydraulic Research*, vol. 49, no. 3, pp. 293–306, 2011.

[13] J. Morison, J. Johnson, and S. Schaaf, "The force exerted by surface waves on piles," *Journal of Petroleum Technology*, vol. 2, pp. 149–154, 05 1950.